LAB 11: Partial Differential Equation

Objective: To numerically obtain the governing PDE of any physical system and solve it numerically.

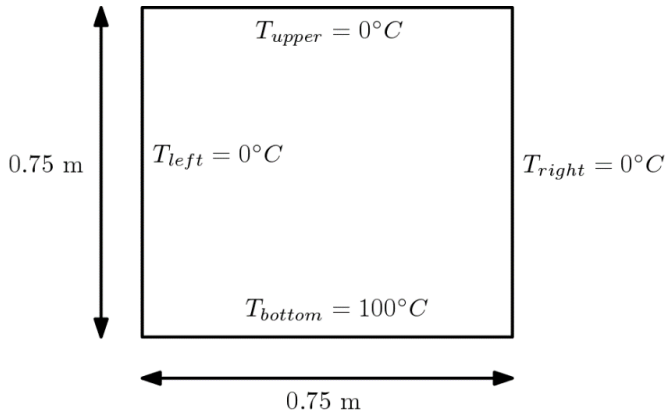
Theory:

Theory - Consider a 2D metal plate with dimensions 0.75x0.75 m2 is kept at 100 degree C at one of it end and rest of the ends are initially at 0 degree C as shown in the figure.

The transient Fourier law of heat conduction for 2D metal plate is given by,

$$\frac{\partial T}{\partial t} = k\left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2}\right)$$

Where T(x,y,t) is the temperature at any location and time and x, y, t are space and time coordinates. Assume the thermal conductivity of plate is K = 1 W/m degree C.



Take the spatial discretization as delta x = delta y = 0.025 m.

Using explicit finite difference method, the discrete equation can be obtained using Taylor series expansion as below,

$$\frac{T_{i,j}^{n+1} - T_{i,j}^{n}}{\Delta t} = k\left(\frac{T_{i+1,j}^{n} - 2T_{i,j}^{n} + T_{i-1,j}^{n}}{\Delta x^2} + \frac{T_{i,j+1}^{n} - 2T_{i,j}^{n} + T_{i,j-1}^{n}}{\Delta y^2}\right)$$

Suppose $\Delta x = \Delta y = \Delta h$

$$T_{i,j}^{n+1} = T_{i,j}^{n} + \frac{k\Delta t}{\Delta h^2}(T_{i+1,j}^{n} - 2T_{i,j}^{n} + T_{i-1,j}^{n} + T_{i,j+1}^{n} - 2T_{i,j}^{n} + T_{i,j-1}^{n})$$

Take the spatial discretization as $\Delta x = \Delta y = \Delta h$ = 0.025 m and $\Delta t = 0.00015$, thermal conductivity of plate is K = 1 W/m degree C.

MATLAB program-

```
% Explicit method for 2D heat equation in flat plate
close all; clear all; clc;
L =0.75;
dx = 0.05;
```

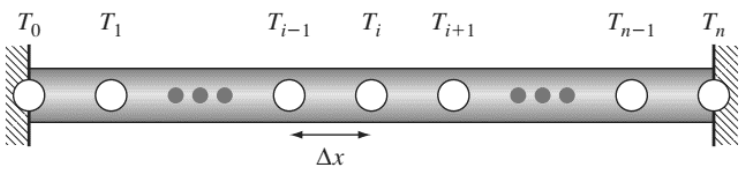
```
dy=0.05;
N = L/dx +1;
x = linspace(0,L,N);
y = linspace(0,L,N);
dt = 0.00015;
epsilon = 1e-4;
T_new = zeros(N,N);
 %boundary condition
      T_new(1,:) =100;
      T_new(:,1) =0;
      T_new(N,:)=0;
      T_new(:,N)=0;

error =1;iter=0;
while(error > epsilon)
    iter=iter+1;
      T = T_new;% update T every iteration
   for i=2:N-1
      for j =2:N-1
         T_new(i,j) = dt*((T(i+1,j)-2*T(i,j)+T(i-1,j))/dx^2 +(T(i,j+1)-2*T(i,j)+T(i,j-1))/dy^2) +
T(i,j);

      end
   end
   error = max(max(abs(T-T_new)));

figure(2);
hold on;
pause(0.001);
%     contourf(T_new);
    shading flat;colorbar;
    contourf(x,y,T_new,'ShowText','on')
colorbar;
xlabel('x meter');ylabel('y meter')

end
```

Assignment -

One numerical as suggested by Faculty.

Sample numerical.

Consider 1D heat conduction in metal rod as below.



1D heat conduction governing equation is given below,

$$\frac{\partial T}{\partial t} = k\left(\frac{\partial^2 T}{\partial x^2}\right)$$

Write down the discretized equation is explicit finite difference method and obtain the Temperature profile T(x,t). Take length of rod is 0.75 m, Δx= 0.025 m, $\Delta t = 0.00015$, thermal conductivity of plate is K = 1 W/m degree C.

Given :

$$\frac{\partial T}{\partial t} = k\frac{\partial^2 T}{\partial x^2}$$

$$\frac{\partial T}{\partial t} = \frac{T^{n+1} - T^n}{\Delta t} \quad \text{——— (1)}$$

$$\frac{\partial^2 T}{\partial x^2} = \frac{T_{i+1}^n - 2T_i^n + T_{i-1}^n}{\Delta x^2} \quad \text{——— (2)}$$

$\therefore$ Substituting (1) and (2) in the initial equation

$$\left[T_i^{n+1} = T_i^n + \frac{k\,\Delta t}{\Delta x^2}\left(T_{i+1}^n + 2T_i^n + T_{i-1}^n\right) \right]$$

↳ Applying this to the code

Code :

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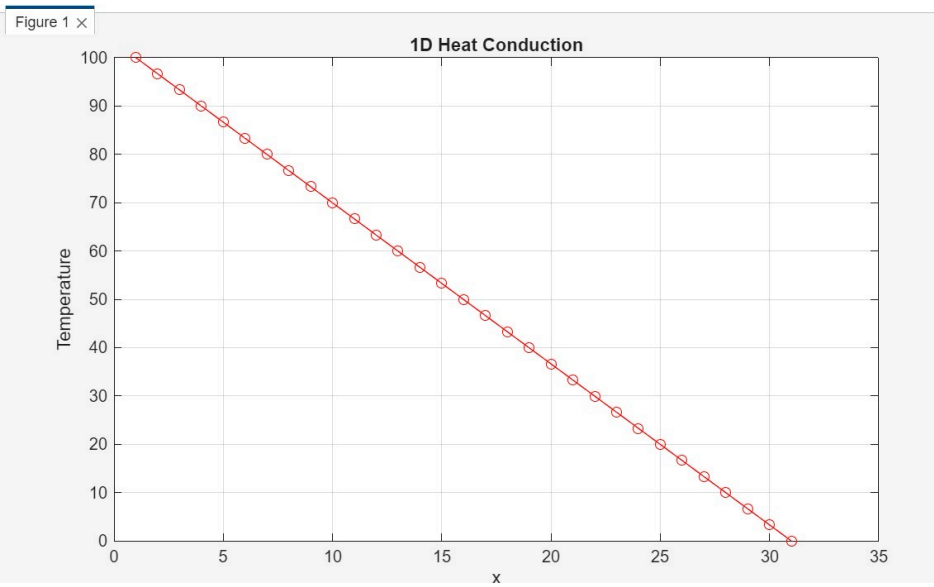
```
% Parameters
L = 0.75; dx = 0.025; dt = 0.00015;
K = 1; eps = 1e-4;
N = L/dx + 1;

% Initialization
T = zeros(1,N);
T(1) = 100;   % left end
T(N) = 0;     % right end

error = 1;

while error > eps
    T_old = T;

    for i = 2:N-1
        T(i) = T_old(i) + K*dt/dx^2 * ...
               (T_old(i+1) - 2*T_old(i) + T_old(i-1));
    end

    error = max(abs(T - T_old));
end

plot(T,'r-o'); grid on;
title('1D Heat Conduction');
xlabel('x'); ylabel('Temperature');
```

→ Applying formula to the code

Result Profile :